



Mobile Application Development

(3161612)

LABORATORY MANUAL

B.E. Semester-6

**Prepared By:-
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5	Write android program to perform all operation in calculator.
6	Write android program to create multiple activities within an application.
7	Write android program to demonstrate action button by implementing ONCLICKLISTENER.
8	Write android program to demonstrate sound button application.
9	Write an Android application to convert into different currencies for example, Rupees to dollar.
10	Write an android application to count library overdue.
11	Write an android application to convert a ball from size of radius 2(colour red) to radius 4(colour blue) to radius 6 (colour green). The ball must rotate in circle for 1 minute before changing size and colour.
12	Write an application to mark the daily route of travel in map.
13	Write an application to record video and audio on topic "Intent" and play the audio and video
14	Write android program to set the wallpaper of your device using bitmap class.

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Software Requirements

Sr No	Software Requirement	Hardware Requirement
1	Android Studio	64-bit OS
		RAM-16 GB
		Processor (Intel i5/i7, AMD Ryzen, etc.)
		Hard Disk- 512 GB



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PRACTICAL 1

AIM: Write Android program to display "Welcome to android".

```
<!-- res/layout/activity_main.xml -->
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">
    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to android"
        android:textSize="24sp"
        android:textStyle="bold"/>
</LinearLayout>
```

Output:



Fig 1.1 Output

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Logical Application

1. What is the default layout file used in an Android activity?

2. Which XML attribute sets the text displayed in a TextView?



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PRACTICAL 2

AIM: Write android program to demonstrate usage of string.

Program:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp"
    android:background="#FFFFFF">

    <TextView
        android:id="@+id/textViewResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="String Example"
        android:textSize="22sp"
        android:textColor="#000000" />
</LinearLayout>
```

Main Activity.java

```
package com.example.stringdemo;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        TextView textView = findViewById(R.id.textViewResult);
```

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```
// String usage demonstration
String firstName = "Android";
String lastName = "Programming";

// Concatenation
String fullName = firstName + " " + lastName;

// Display result
textView.setText("Concatenated String: " + fullName);
}
```

Output:

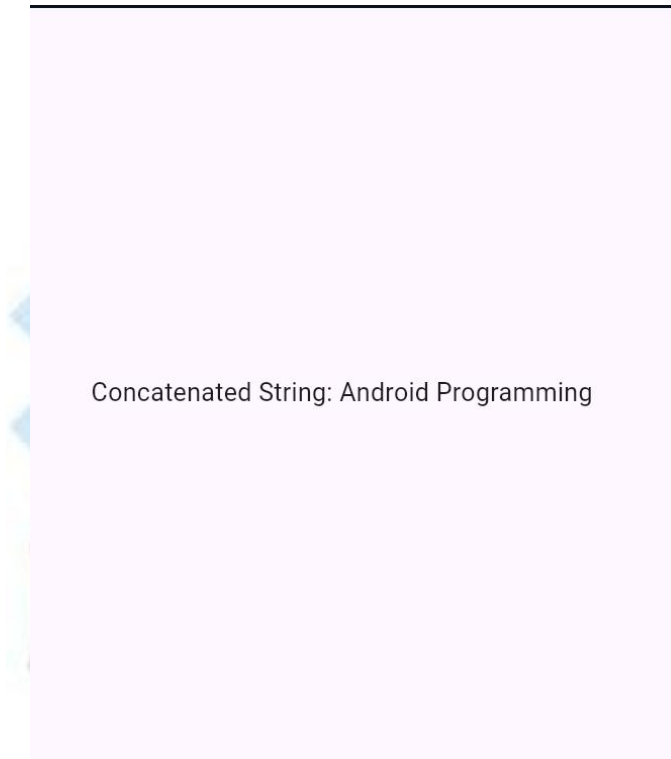


Fig 2.1 Output

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Logical Application

1. What class in Android is used to store string resources?

2. How do you access a string resource in Java using its ID?



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PRACTICAL 3

AIM: Write android program to demonstrate activity life cycle.

Program:

XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:background="#FFFFFF"
    android:padding="20dp">

    <TextView
        android:id="@+id/textViewLifeCycle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Activity Life Cycle Demo"
        android:textSize="20sp"
        android:textColor="#000000"/>
</LinearLayout>
```

Java Code (MainActivity.java)

```
package com.example.lifecycleapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.util.Log;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {
    private static final String TAG = "ActivityLifeCycle";
    TextView textView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        textView = findViewById(R.id.textViewLifeCycle);
        textView.setText("onCreate() called");
        Log.d(TAG, "onCreate() called");
    }
}
```

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```
@Override
protected void onStart() {
    super.onStart();
    Log.d(TAG, "onStart() called");
}

@Override
protected void onResume() {
    super.onResume();
    Log.d(TAG, "onResume() called");
}

@Override
protected void onPause() {
    super.onPause();
    Log.d(TAG, "onPause() called");
}

@Override
protected void onStop() {
    super.onStop();
    Log.d(TAG, "onStop() called");
}

@Override
protected void onRestart() {
    super.onRestart();
    Log.d(TAG, "onRestart() called");
}

@Override
protected void onDestroy() {
    super.onDestroy();
    Log.d(TAG, "onDestroy() called");
}
}
```

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Output:



Fig 3.1 Output

Logical Application

1. Which method is used to set the background color of a layout programmatically?

2. What XML attribute sets the background color of a View?

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PRACTICAL 4

AIM: Write android program to change the background of activity.

Program:

XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp"
    android:id="@+id/mainLayout">

    <Button
        android:id="@+id/btnChangeColor"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Change Background" />

</LinearLayout>
```

Java Code (MainActivity.java)

```
package com.example.backgrounddemo;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.graphics.Color;
import android.view.View;
import android.widget.Button;
import android.widget.LinearLayout;
public class MainActivity extends AppCompatActivity {
    LinearLayout mainLayout;
    Button btnChange;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        mainLayout = findViewById(R.id.mainLayout);
        btnChange = findViewById(R.id.btnChangeColor);
        btnChange.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Change background color on button click
                mainLayout.setBackgroundColor(Color.CYAN);
            }
        })
    }
}
```

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```
    });  
  }  
}
```

Output :

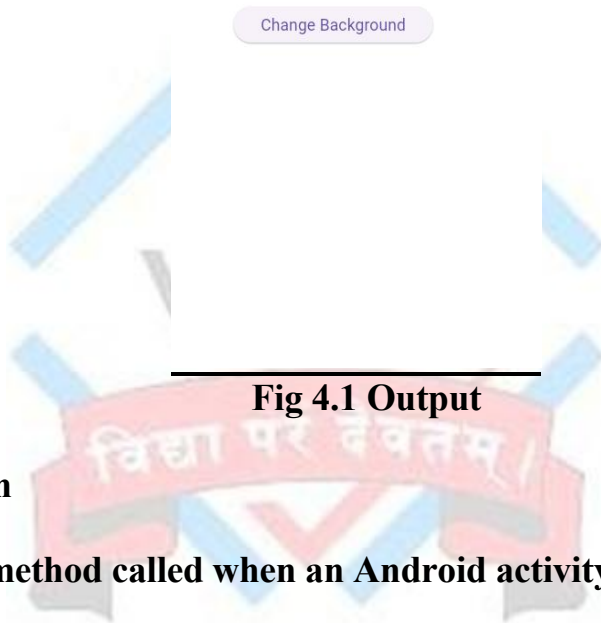


Fig 4.1 Output

Logical Application

1. Name the first method called when an Android activity is created.

2. Which lifecycle method is called when an activity becomes invisible to the user?

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PRACTICAL 5

AIM: Write android program to perform all operation in calculator.

Program:

XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center"
    android:id="@+id/mainLayout">

    <EditText
        android:id="@+id/etNum1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter First Number"
        android:inputType="numberDecimal"
        android:layout_marginBottom="10dp"/>

    <EditText
        android:id="@+id/etNum2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Second Number"
        android:inputType="numberDecimal"
        android:layout_marginBottom="20dp"/>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:gravity="center"
        android:layout_marginBottom="20dp">

        <Button
            android:id="@+id/btnAdd"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="+" />
```

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```
<Button
    android:id="@+id/btnSub"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="-"
    android:layout_marginLeft="10dp"/>
```

```
<Button
    android:id="@+id/btnMul"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="×"
    android:layout_marginLeft="10dp"/>
```

```
<Button
    android:id="@+id/btnDiv"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="÷"
    android:layout_marginLeft="10dp"/>
```

```
</LinearLayout>
```

```
<TextView
    android:id="@+id/tvResult"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Result will be shown here"
    android:textSize="20sp"
    android:textColor="#000000"/>
```

```
</LinearLayout>
```

Java Code (MainActivity.java)

```
package com.example.calculatordemo;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
```

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```
EditText etNum1, etNum2;  
Button btnAdd, btnSub, btnMul, btnDiv;  
TextView tvResult;
```

@Override

```
protected void onCreate(Bundle savedInstanceState) {  
    super.onCreate(savedInstanceState);  
    setContentView(R.layout.activity_main);
```

```
    etNum1 = findViewById(R.id.etNum1);  
    etNum2 = findViewById(R.id.etNum2);  
    btnAdd = findViewById(R.id.btnAdd);  
    btnSub = findViewById(R.id.btnSub);  
    btnMul = findViewById(R.id.btnMul);  
    btnDiv = findViewById(R.id.btnDiv);  
    tvResult = findViewById(R.id.tvResult);
```

```
    btnAdd.setOnClickListener(new View.OnClickListener() {  
        @Override  
        public void onClick(View v) {  
            calculate('+');  
        }  
    });
```

```
    btnSub.setOnClickListener(new View.OnClickListener() {  
        @Override  
        public void onClick(View v) {  
            calculate('-');  
        }  
    });
```

```
    btnMul.setOnClickListener(new View.OnClickListener() {  
        @Override  
        public void onClick(View v) {  
            calculate('*');  
        }  
    });
```

```
    btnDiv.setOnClickListener(new View.OnClickListener() {  
        @Override  
        public void onClick(View v) {  
            calculate('/');  
        }  
    });
```


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```
}

private void calculate(char operator) {
    String num1Str = etNum1.getText().toString();
    String num2Str = etNum2.getText().toString();

    if (num1Str.isEmpty() || num2Str.isEmpty()) {
        Toast.makeText(this, "Please enter both numbers", Toast.LENGTH_SHORT).show();
        return;
    }

    double num1 = Double.parseDouble(num1Str);
    double num2 = Double.parseDouble(num2Str);
    double result = 0;

    switch (operator) {
        case '+':
            result = num1 + num2;
            break;
        case '-':
            result = num1 - num2;
            break;
        case '*':
            result = num1 * num2;
            break;
        case '/':
            if (num2 != 0) {
                result = num1 / num2;
            } else {
                Toast.makeText(this, "Cannot divide by zero", Toast.LENGTH_SHORT).show();
                return;
            }
            break;
    }

    tvResult.setText("Result: " + result);
}
}
```

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OUTPUT :

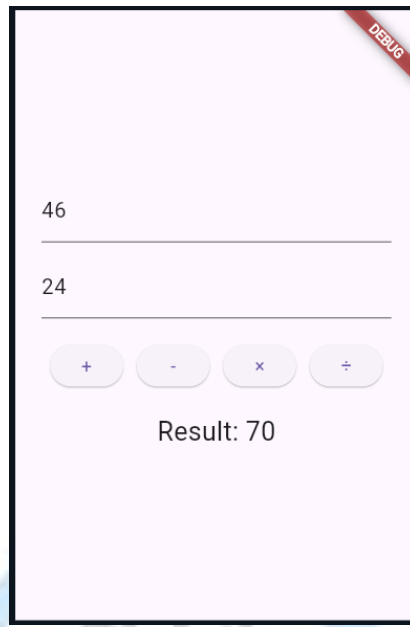


Fig 5.1 Output

Logical Application

1. Which listener is used to handle button clicks in a calculator app?

2. What data type is best suited for storing decimal calculator results in Java?

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Date:

PRACTICAL 6

AIM: Write android program to create multiple activities within an application.

Program:

XML Layout for MainActivity (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <Button
        android:id="@+id/btnNext"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Go to Second Activity" />
</LinearLayout>
```

XML Layout for SecondActivity (activity_second.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <TextView
        android:id="@+id/tvMessage"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to Second Activity"
        android:textSize="20sp"
        android:textColor="#000000"/>
</LinearLayout>
```

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Java Code for MainActivity.java

```
package com.example.multipleactivities;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
public class MainActivity extends AppCompatActivity {
    Button btnNext;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnNext = findViewById(R.id.btnNext);

        btnNext.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Intent to open SecondActivity
                Intent intent = new Intent(MainActivity.this, SecondActivity.class);
                startActivity(intent);
            }
        });
    }
}
```

Java Code for SecondActivity.java

```
package com.example.multipleactivities;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class SecondActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);
    }
}
```

AndroidManifest.xml (Add Second Activity)

```
<application
    android:allowBackup="true"
    android:icon="@mipmap/ic_launcher"
```

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```
android:label="@string/app_name"  
android:supportsRtl="true"  
android:theme="@style/Theme.AppCompat.Light.NoActionBar">  
  
<activity android:name=".SecondActivity"></activity>  
<activity android:name=".MainActivity">  
    <intent-filter>  
        <action android:name="android.intent.action.MAIN" />  
        <category android:name="android.intent.category.LAUNCHER" />  
    </intent-filter>  
</activity>  
</application>
```

Output :

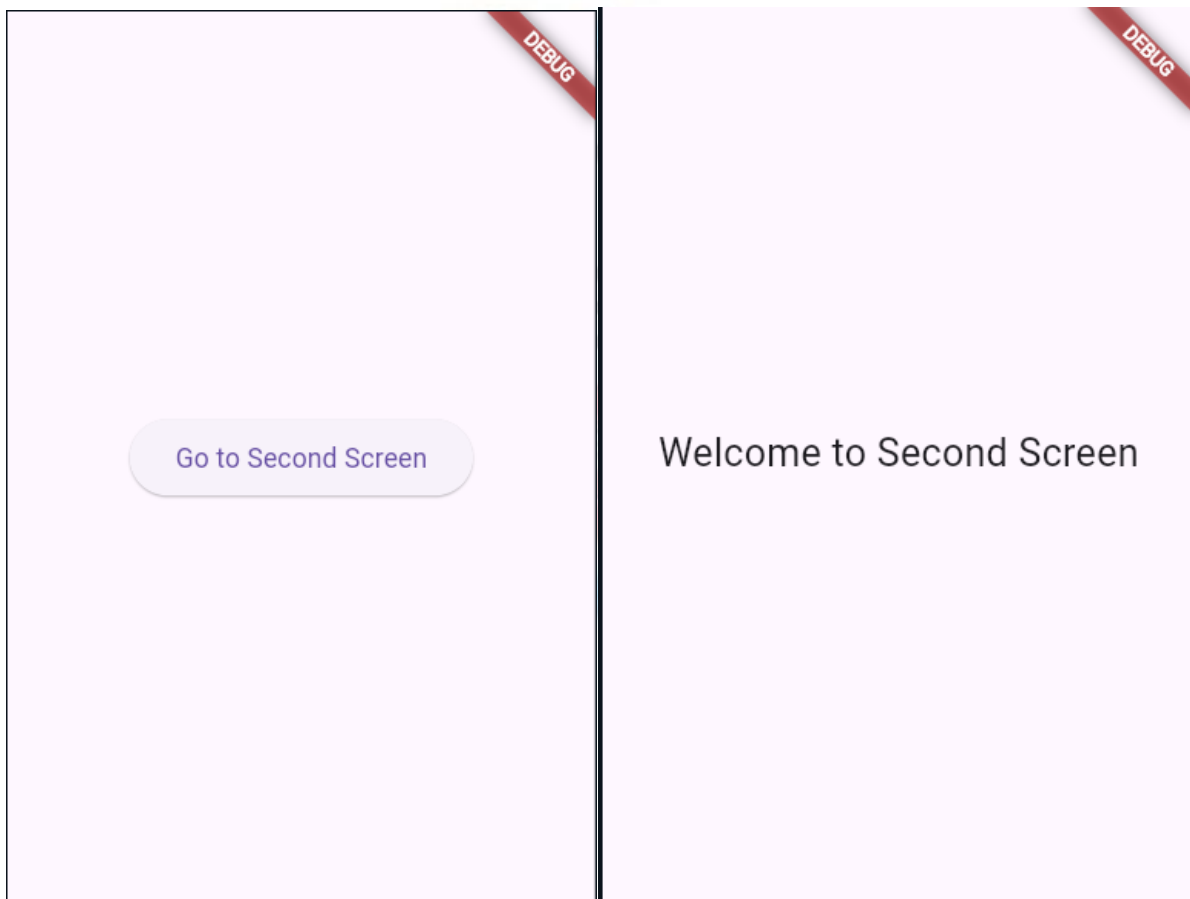


Fig 6.1 Output

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Logical Application

1. Which class is used to navigate from one activity to another in Android?

2. What method is called to start a new activity using an Intent?



Date:

PRACTICAL 7

AIM: Write android program to demonstrate action button by implementing ONCLICKLISTENER.

Program:

XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <Button
        android:id="@+id/btnClickMe"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click Me" />

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Result will appear here"
        android:textSize="20sp"
        android:layout_marginTop="20dp"/>
</LinearLayout>
```

Java Code (MainActivity.java)

```
package com.example.onclickdemo;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity implements View.OnClickListener {

    Button btnClickMe;
    TextView tvResult;
```

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```
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    btnClickMe = findViewById(R.id.btnClickMe);
    tvResult = findViewById(R.id.tvResult);

    // Set OnClickListener
    btnClickMe.setOnClickListener(this);
}

@Override
public void onClick(View v) {
    if (v.getId() == R.id.btnClickMe) {
        tvResult.setText("Button Clicked Successfully!");
    }
}
```

Output :

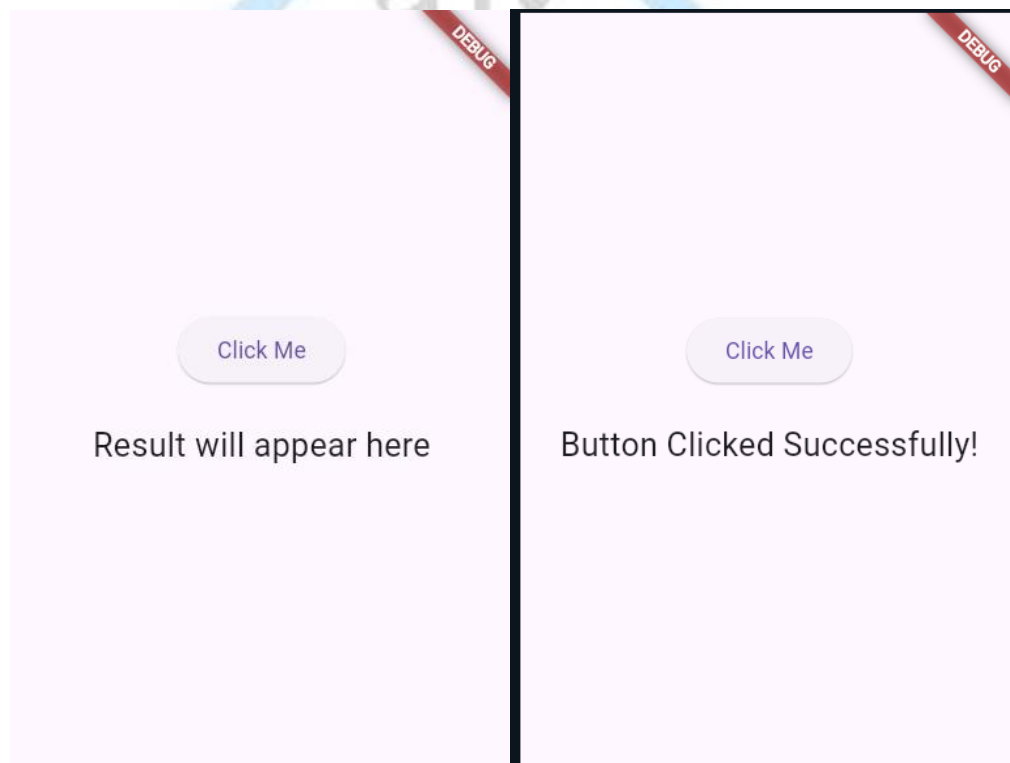


Fig 7.1 Output

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Logical Application

1. Which interface must be implemented to handle button click events?

2. What method do you override when implementing OnClickListener?



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PRACTICAL 8

AIM: Write android program to demonstrate sound button application.

Program:

XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <Button
        android:id="@+id/btnPlaySound"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Play Sound" />
</LinearLayout>
```

Java Code (MainActivity.java)

```
package com.example.soundbuttonapp;
import androidx.appcompat.app.AppCompatActivity;
import android.media.MediaPlayer;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {

    Button btnPlaySound;
    MediaPlayer mediaPlayer;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btnPlaySound = findViewById(R.id.btnPlaySound);

        // Initialize MediaPlayer with sound file from res/raw
        mediaPlayer = MediaPlayer.create(this, R.raw.clicksound);
        btnPlaySound.setOnClickListener(new View.OnClickListener() {
```

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```
@Override
public void onClick(View v) {
    // Play sound when button clicked
    if (mediaPlayer != null) {
        mediaPlayer.start();
    }
}

@Override
protected void onDestroy() {
    super.onDestroy();
    if (mediaPlayer != null) {
        mediaPlayer.release();
        mediaPlayer = null;
    }
}
```

Output :

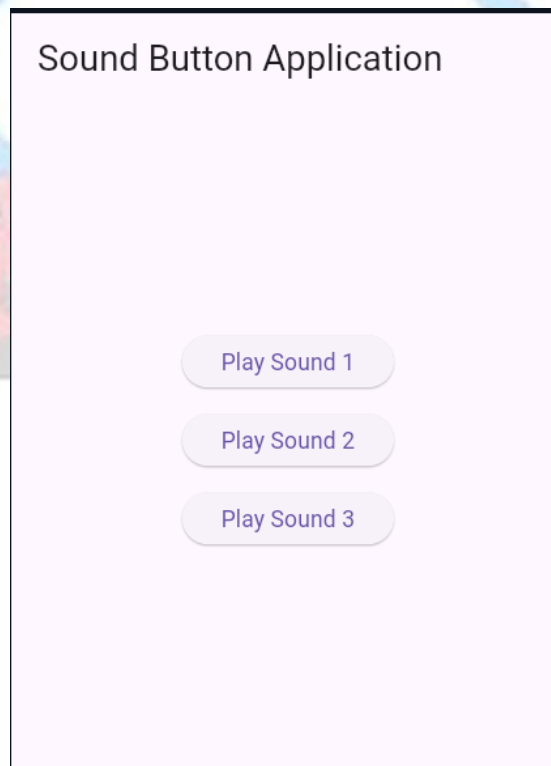


Fig 8.1 Output

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Logical Application

1. Which Android class is used to play short sound effects?

2. What method of MediaPlayer is called to begin audio playback?



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PRACTICAL 9

AIM: Write an Android application to convert into different currencies for example, Rupees to dollar.

Program:

XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <EditText
        android:id="@+id/etRupees"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter amount in Rupees"
        android:inputType="numberDecimal"
        android:layout_marginBottom="15dp"/>

    <Button
        android:id="@+id/btnConvert"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Convert to Dollar" />

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Result will appear here"
        android:textSize="20sp"
        android:layout_marginTop="20dp"/>
</LinearLayout>
```

Java Code (MainActivity.java)

```
package com.example.currencyconverter;
import androidx.appcompat.app.AppCompatActivity;
```

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```
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    EditText etRupees;
    Button btnConvert;
    TextView tvResult;

    // Conversion rate (Example: 1 Rupee = 0.012 USD)
    final double DOLLAR_RATE = 0.012;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etRupees = findViewById(R.id.etRupees);
        btnConvert = findViewById(R.id.btnConvert);
        tvResult = findViewById(R.id.tvResult);

        btnConvert.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String rupeesStr = etRupees.getText().toString();

                if (rupeesStr.isEmpty()) {
                    Toast.makeText(MainActivity.this, "Please enter amount in Rupees",
Toast.LENGTH_SHORT).show();
                    return;
                }

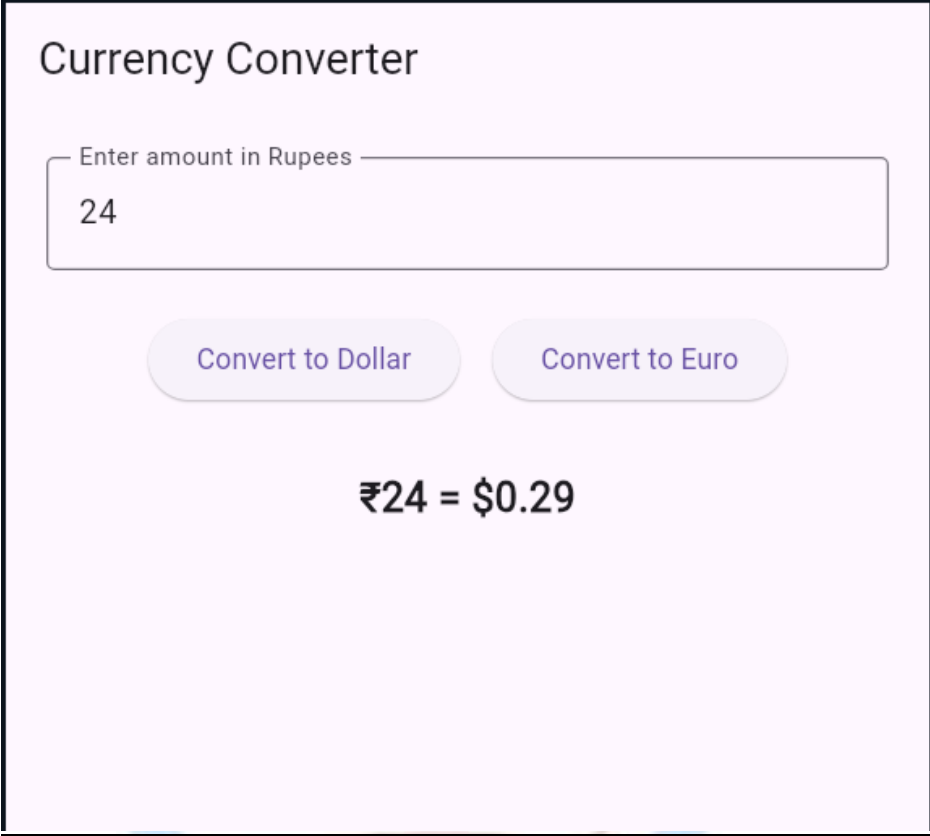
                double rupees = Double.parseDouble(rupeesStr);
                double dollars = rupees * DOLLAR_RATE;

                tvResult.setText("USD: $" + String.format("%.2f", dollars));
            }
        });
    }
}
```

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Output :



The screenshot shows a mobile application titled "Currency Converter". It features a text input field with the placeholder text "Enter amount in Rupees" and the value "24". Below the input field are two buttons: "Convert to Dollar" and "Convert to Euro". The result of the conversion is displayed as "₹24 = \$0.29".

Fig 9.1 Output

Logical Application

1. What layout widget is used to take numeric input from the user?

2. How do you convert a String from an EditText to a float in Java?

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Date:

PRACTICAL 10

AIM: Write an android application to count library overdue.

Program:

XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <EditText
        android:id="@+id/etDays"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter number of overdue days"
        android:inputType="number"
        android:layout_marginBottom="15dp"/>

    <Button
        android:id="@+id/btnCalculate"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Calculate Fine" />

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Fine will appear here"
        android:textSize="20sp"
        android:layout_marginTop="20dp"/>
</LinearLayout>
```


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Java Code (MainActivity.java)

```
package com.example.libraryfine;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity {

    EditText etDays;
    Button btnCalculate;
    TextView tvResult;

    // Fine rate per day
    final int FINE_RATE = 2;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etDays = findViewById(R.id.etDays);
        btnCalculate = findViewById(R.id.btnCalculate);
        tvResult = findViewById(R.id.tvResult);

        btnCalculate.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String daysStr = etDays.getText().toString();

                if (daysStr.isEmpty()) {
                    Toast.makeText(MainActivity.this, "Please enter overdue days",
Toast.LENGTH_SHORT).show();
                    return;
                }
                int days = Integer.parseInt(daysStr);
                int fine = days * FINE_RATE;

                tvResult.setText("Total Fine: ₹" + fine);
            }
        });
    }
}
```

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Output :

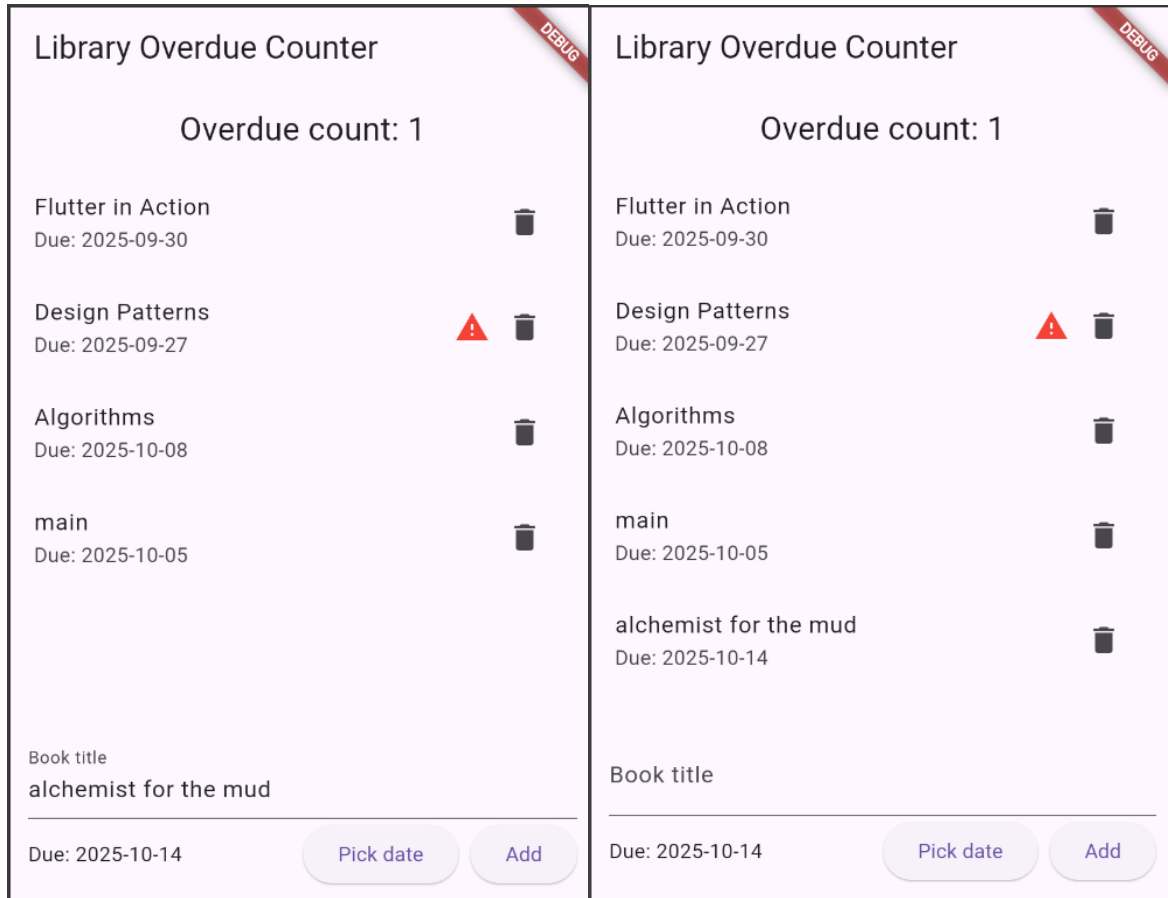


Fig 10.1 Output

Logical Application

1. Which Java class is used to calculate the difference between two dates?

2. What widget is used to let the user pick a date in Android?

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Date:

PRACTICAL 11

AIM: Write an application to mark the daily route of travel in map.

Program:

XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<fragment xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/map"
    android:name="com.google.android.gms.maps.SupportMapFragment"
    android:layout_width="match_parent"
    android:layout_height="match_parent"/>
```

Java Code (MainActivity.java)

```
package com.example.dailyroute;
import androidx.fragment.app.FragmentActivity;
import android.os.Bundle;
import com.google.android.gms.maps.CameraUpdateFactory;
import com.google.android.gms.maps.GoogleMap;
import com.google.android.gms.maps.OnMapReadyCallback;
import com.google.android.gms.maps.SupportMapFragment;
import com.google.android.gms.maps.model.LatLng;
import com.google.android.gms.maps.model.MarkerOptions;
import com.google.android.gms.maps.model.PolylineOptions;
public class MainActivity extends FragmentActivity implements OnMapReadyCallback {
    private GoogleMap mMap;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        SupportMapFragment mapFragment = (SupportMapFragment) getSupportFragmentManager()
            .findFragmentById(R.id.map);
        if (mapFragment != null) {
            mapFragment.getMapAsync(this);
        }
    }
    @Override
    public void onMapReady(GoogleMap googleMap) {
        mMap = googleMap;

        // Example route: Home -> Bus Stop -> Office
        LatLng home = new LatLng(28.6139, 77.2090); // Delhi example
```

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```
LatLng busStop = new LatLng(28.6200, 77.2100);
LatLng office = new LatLng(28.6250, 77.2150);
// Add markers
mMap.addMarker(new MarkerOptions().position(home).title("Home"));
mMap.addMarker(new MarkerOptions().position(busStop).title("Bus Stop"));
mMap.addMarker(new MarkerOptions().position(office).title("Office"));
// Draw route with Polyline
mMap.addPolyline(new PolylineOptions()
    .add(home, busStop, office)
    .width(8)
    .color(0xFF2196F3)); // Blue
// Move camera to Home
mMap.moveCamera(CameraUpdateFactory.newLatLngZoom(home, 14));
}
```

Output :

Daily Travel Route	
Enter stop name	Enter stop name
st. sebastian trail	
Add Stop	Add Stop
Clear Route	Clear Route
1 malteasers tower	1 malteasers tower
	2 st. sebastian trail

Fig 11.1 Output

Logical Application

1. Which Android class is used to create animations on a View?

2. What method changes the color of a shape drawable programmatically?



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Date:

PRACTICAL 12

AIM: Write an Android application to convert into different currencies.

Program:

XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <EditText
        android:id="@+id/etRupees"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter amount in Rupees"
        android:inputType="numberDecimal"
        android:layout_marginBottom="15dp"/>

    <Button
        android:id="@+id/btnConvert"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Convert to Dollar" />

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Result will appear here"
        android:textSize="20sp"
        android:layout_marginTop="20dp"/>
</LinearLayout>
```

Java Code (MainActivity.java)

```
package com.example.currencyconverter;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
```

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```
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    EditText etRupees;
    Button btnConvert;
    TextView tvResult;

    // Conversion rate (Example: 1 Rupee = 0.012 USD)
    final double DOLLAR_RATE = 0.012;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etRupees = findViewById(R.id.etRupees);
        btnConvert = findViewById(R.id.btnConvert);
        tvResult = findViewById(R.id.tvResult);

        btnConvert.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String rupeesStr = etRupees.getText().toString();

                if (rupeesStr.isEmpty()) {
                    Toast.makeText(MainActivity.this, "Please enter amount in Rupees",
Toast.LENGTH_SHORT).show();
                    return;
                }

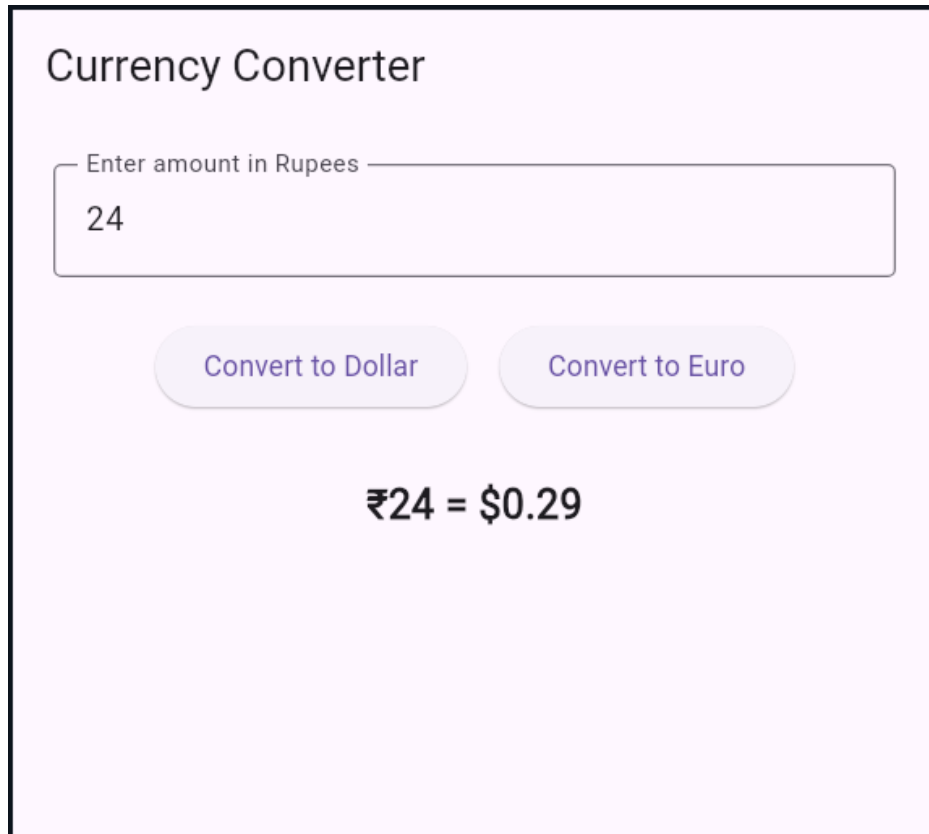
                double rupees = Double.parseDouble(rupeesStr);
                double dollars = rupees * DOLLAR_RATE;

                tvResult.setText("USD: $" + String.format("%.2f", dollars));
            }
        });
    }
}
```

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Output:



Currency Converter

Enter amount in Rupees

24

Convert to Dollar

Convert to Euro

₹24 = \$0.29

Fig 12.1 Output

Logical Application

1. Which Google Maps API method is used to draw a path on a map?

2. What permission is required in AndroidManifest to access the device location?

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Date:

PRACTICAL 13

AIM: Write an application to record video and audio on topic "Intent" and play the audio and video.

Program:

AndroidManifest.xml

```
<uses-permission android:name="android.permission.CAMERA"/>
<uses-permission android:name="android.permission.RECORD_AUDIO"/>
<uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>
<uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE"/>

<application
    ...>
    <activity android:name=".MainActivity">
        <intent-filter>
            <action android:name="android.intent.action.MAIN"/>
            <category android:name="android.intent.category.LAUNCHER"/>
        </intent-filter>
    </activity>
</application>
```

XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp">

    <Button
        android:id="@+id/btnRecordAudio"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Record Audio"/>

    <Button
        android:id="@+id/btnRecordVideo"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Record Video"
        android:layout_marginTop="15dp"/>
```

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```
<Button
    android:id="@+id/btnPlayAudio"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Play Audio"
    android:layout_marginTop="15dp"/>

<Button
    android:id="@+id/btnPlayVideo"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Play Video"
    android:layout_marginTop="15dp"/>
</LinearLayout>
```

Java Code (MainActivity.java)

```
package com.example.intentmediaapp;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.provider.MediaStore;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    Button btnRecordAudio, btnRecordVideo, btnPlayAudio, btnPlayVideo;
    Uri audioUri, videoUri;
    final int AUDIO_REQUEST = 1;
    final int VIDEO_REQUEST = 2;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnRecordAudio = findViewById(R.id.btnRecordAudio);
        btnRecordVideo = findViewById(R.id.btnRecordVideo);
        btnPlayAudio = findViewById(R.id.btnPlayAudio);
        btnPlayVideo = findViewById(R.id.btnPlayVideo);
    }
}
```

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```
// Record Audio
btnRecordAudio.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {

        Intent audioIntent = new Intent(MediaStore.Audio.Media.RECORD_SOUND_ACTION);
        try {
            startActivityForResult(audioIntent, AUDIO_REQUEST);
        } catch (Exception e) {
            Toast.makeText(MainActivity.this, "No app found to record audio",
Toast.LENGTH_SHORT).show();
        }
    }
});

// Record Video
btnRecordVideo.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {

        Intent videoIntent = new Intent(MediaStore.ACTION_VIDEO_CAPTURE);
        try {
            startActivityForResult(videoIntent, VIDEO_REQUEST);
        } catch (Exception e) {
            Toast.makeText(MainActivity.this, "No app found to record video",
Toast.LENGTH_SHORT).show();
        }
    }
});

// Play Audio
btnPlayAudio.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        if (audioUri != null) {
            Intent playAudio = new Intent(Intent.ACTION_VIEW);
            playAudio.setDataAndType(audioUri, "audio/*");
            startActivity(playAudio);
        } else {
            Toast.makeText(MainActivity.this, "No audio recorded", Toast.LENGTH_SHORT).show();
        }
    }
});

// Play Video
```

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```
btnPlayVideo.setOnClickListener(new View.OnClickListener() {  
    @Override  
    public void onClick(View v) {  
        if (videoUri != null) {  
            Intent playVideo = new Intent(Intent.ACTION_VIEW);  
            playVideo.setDataAndType(videoUri, "video/*");  
            startActivity(playVideo);  
        } else {  
            Toast.makeText(MainActivity.this, "No video recorded", Toast.LENGTH_SHORT).show();  
        }  
    }  
});  
}  
@Override  
protected void onActivityResult(int requestCode, int resultCode, Intent data) {  
    super.onActivityResult(requestCode, resultCode, data);  
    if (resultCode == RESULT_OK) {  
        if (requestCode == AUDIO_REQUEST) {  
            audioUri = data.getData();  
        } else if (requestCode == VIDEO_REQUEST) {  
            videoUri = data.getData();  
        }  
    }  
}
```

Output :

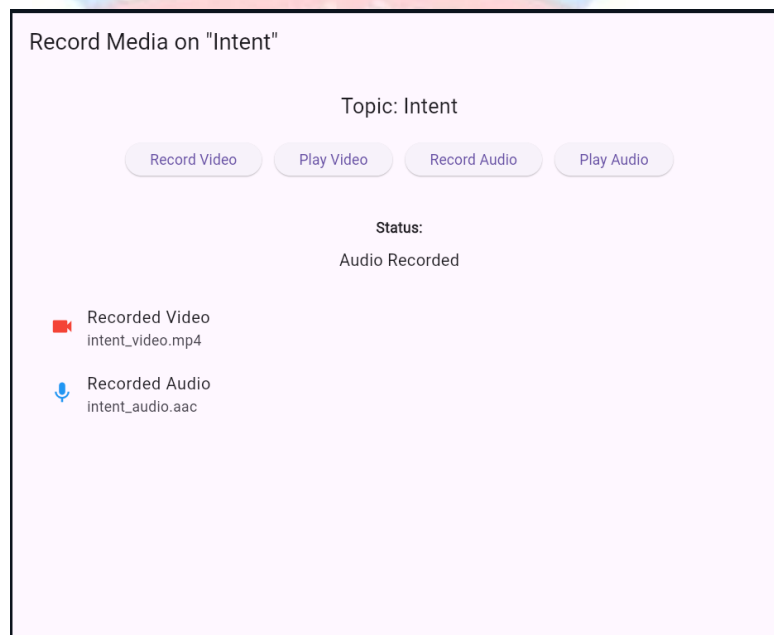


Fig 13.1 Output

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Logical Application

1. Which class in Android is used to record audio and video?

2. What method of MediaPlayer is used to specify the file path of media to play?



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Date:

PRACTICAL 14

AIM: Write android program to set the wallpaper of your device using bitmap class.

Program:

XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp">
    <Button
        android:id="@+id/btnSetWallpaper"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Set Wallpaper"/>
</LinearLayout>
```

Java Code (MainActivity.java)

```
package com.example.wallpaperapp;

import androidx.appcompat.app.AppCompatActivity;
import android.app.WallpaperManager;
import android.graphics.Bitmap;
import android.graphics.BitmapFactory;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;
import java.io.IOException;

public class MainActivity extends AppCompatActivity {
    Button btnSetWallpaper;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btnSetWallpaper = findViewById(R.id.btnSetWallpaper);
        btnSetWallpaper.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
```

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```
// Load bitmap from drawable
Bitmap bitmap = BitmapFactory.decodeResource(getResources(),
R.drawable.wallpaper_image);
// Set wallpaper using WallpaperManager
WallpaperManager wallpaperManager =
WallpaperManager.getInstance(getApplicationContext());
try {
    wallpaperManager.setBitmap(bitmap);
    Toast.makeText(MainActivity.this, "Wallpaper Set Successfully",
Toast.LENGTH_SHORT).show();
} catch (IOException e) {
    e.printStackTrace();
    Toast.makeText(MainActivity.this, "Failed to Set Wallpaper",
Toast.LENGTH_SHORT).show();
}
}
});
}
```

Output :

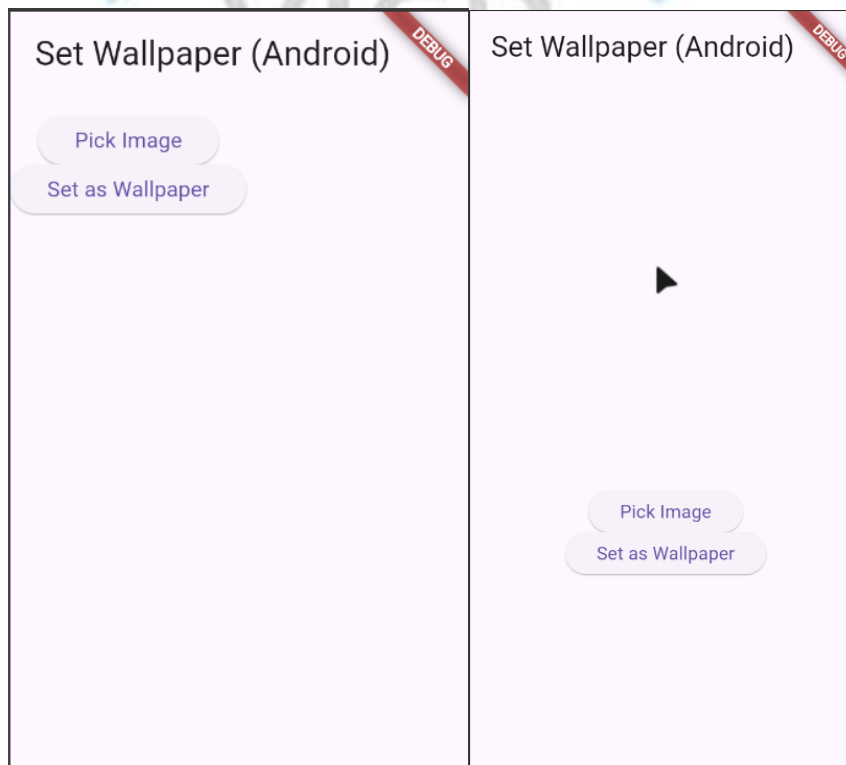


Fig 14.1 Output

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Logical Application

1. Which Android class provides the method to set the device wallpaper?

2. What permission must be declared in the manifest to change the wallpaper?

